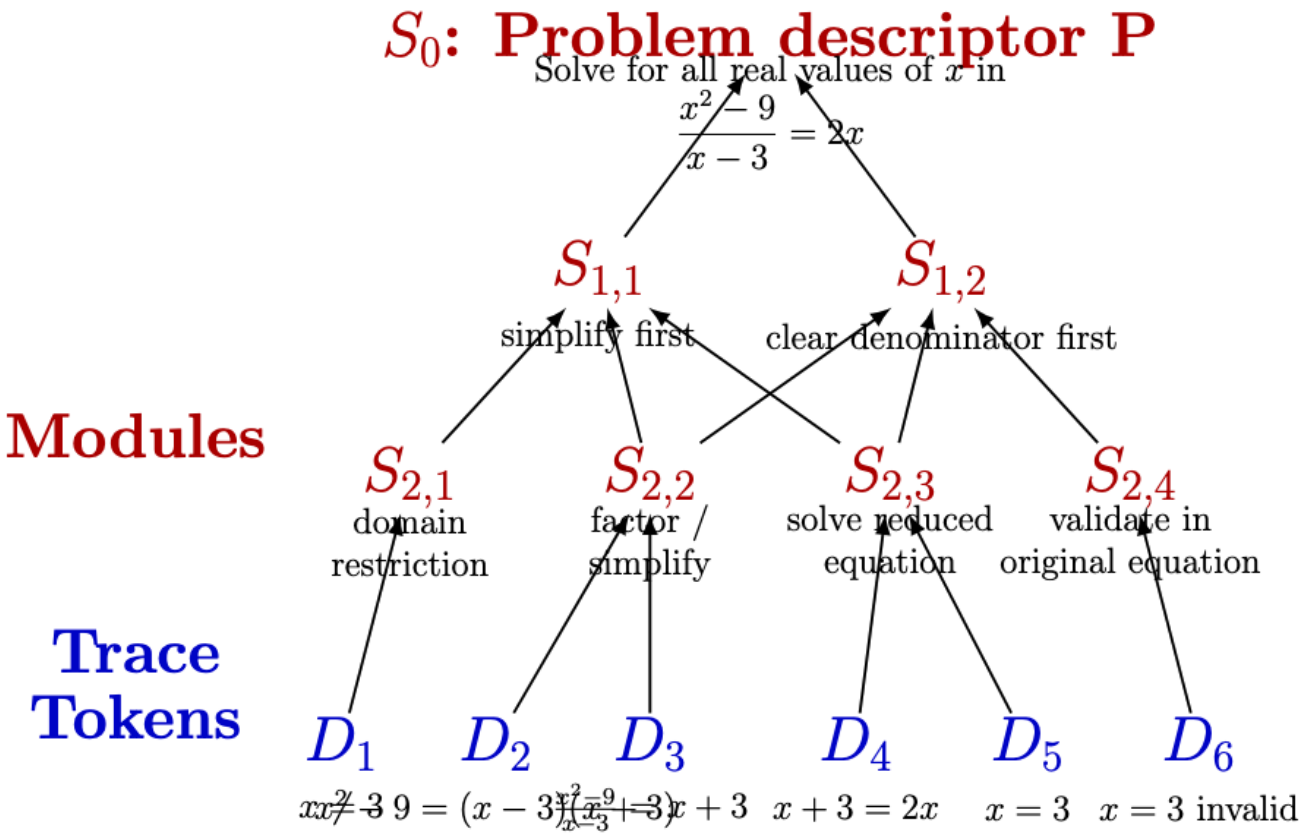

Figure H6. **A concrete algebra example illustrating skills, routing mechanisms, and identification conditions.** xxx.

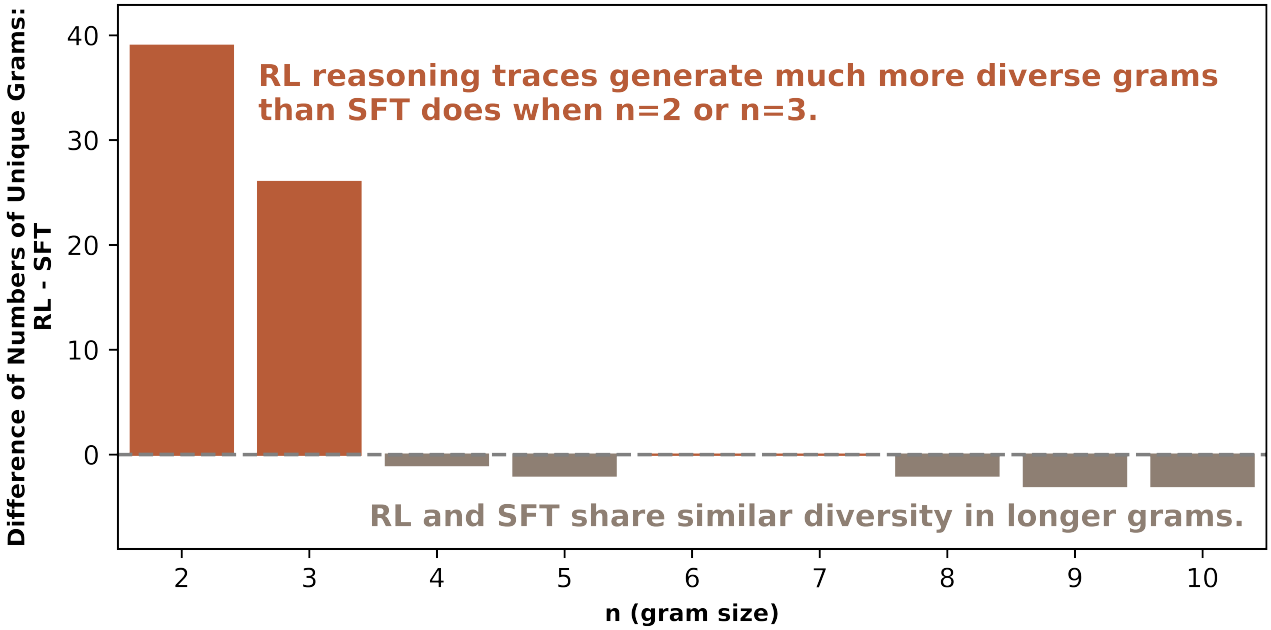

Figure H7. **Difference of numbers of unique grams between the RL and SFT model.** For the n-grams obtained in Appendix D (lines 979-983), we calculate the number of unique grams for both RL and SFT models, then visualize their differences here. The results show that (1) RL reasoning traces generate much more diverse grams than SFT does when $n = 2$ and $n = 3$; (2) RL and SFT reasoning traces share similar diversity in longer grams.